

Replication of Study X by Sample & Sample (20xx, Psychological Science)

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Introduction

I am largely interested in the study of memory from a cognitive perspective, as there is still much to explore about the features and limitations of working memory in all forms. I have previously focused on much more rudimentary, but rather mundane motor skill cognition tasks, which have introduced me to creating my own web experiment with javascript. With the paper I chose being focused on visual working memory, (Brady and Störmer, 2022) I feel sufficiently challenged by coding a completely new paradigm in a format I'm familiar with.

The first experiment in the paper yielded many different findings about the capacity of visual working memory concerning meaningful items when compared to colors in 2 different formats (simultaneous and sequential).

For this task, I will need to reproduce the visual memory task using the standard color circle (Suchow et al. 2013; Schurgin et al. 2018) and particular set of objects (Brady et al., 2016) in the task from the paper I will be replicating. There will be four distinct styles of task that I need to code: simultaneous color wheel, sequential color wheel, simultaneous objects, and sequential objects. For the two conditions (simultaneous vs. sequential) there are distinct durations for how long each stimuli is displayed. This will benefit from being heavily thought out beforehand, as I can foresee issues with accurately mirroring the order of specific items from the original task.

The first task I coded was much simpler than this, and was concerned mainly with precision of recording input, not precision with displaying stimuli, so working with quickly changing stimuli on a screen will be new. I also received a substantial amount of help with the structure of the code in general, debugging, and most of the work in css and html.

Link to Repository: <https://github.com/jamesrashby21/brady2022>

Link to Replication Paper: <https://dx.doi.org/10.1037/xlm0001014>

[No abstract is needed.] Each replication project will have a straightforward, no frills report of the study and results. These reports will be publicly available as supplementary material for the aggregate report(s) of the project as a whole. Also, to maximize project integrity, the intro and methods will be written and critiqued in advance of data collection. Introductions can be just 1-2 paragraphs clarifying the main idea of the original study, the target finding for replication, and any other essential information. You will likely contact the authors of the original study, so this should be professional and forward facing. It will NOT have a literature review – that is in the original publication. You can write both the introduction and the methods in past tense.

Methods

Power Analysis

Original effect size, power analysis for samples to achieve 80%, 90%, 95% power to detect that effect size. Considerations of feasibility for selecting planned sample size.

Planned Sample

Planned sample size and/or termination rule, sampling frame, known demographics if any, preselection rules if any.

Materials

All materials - can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Procedure

Can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Analysis Plan

Can also quote directly, though it is less often spelled out effectively for an analysis strategy section. The key is to report an analysis strategy that is as close to the original - data cleaning rules, data exclusion rules, covariates, etc. - as possible.

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

Differences from Original Study

Explicitly describe known differences in sample, setting, procedure, and analysis plan from original study. The goal, of course, is to minimize those differences, but differences will inevitably occur. Also, note whether such differences are anticipated to make a difference based on claims in the original article or subsequent published research on the conditions for obtaining the effect.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or “none”.

Results**Data preparation**

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion**Summary of Replication Attempt**

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.